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AISI Type 420F(Se) Stainless Steel, annealed bar**Categories:** [Metal](#); [Ferrous Metal](#); [Heat Resisting](#); [Stainless Steel](#); [T 400 Series Stainless Steel](#)**Material****Notes:****Key Words:** UNS S42023, AMS 5620, SAE J405 (51420FSe), martensitic**Vendors:** No vendors are listed for this material. Please [click here](#) if you are a supplier and would like information on how to add your listing to this material.
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Physical Properties	Metric	English	Comments
Density	7.80 g/cc	0.282 lb/in ³	
Mechanical Properties	Metric	English	Comments
Hardness, Brinell	220	220	
Hardness, Rockwell C	18	18	Converted from Brinell Hardness. Value below normal Rockwell C Hardness range, for comparison only.
Tensile Strength, Ultimate	655 MPa	95000 psi	
Tensile Strength, Yield	380 MPa @Strain 0.2 %	55100 psi @Strain 0.2 %	
Elongation at Break	22 %	22 %	in 50 mm
Modulus of Elasticity	200 GPa	29000 ksi	
Electrical Properties	Metric	English	Comments
Electrical Resistivity	0.0000550 ohm-cm @Temperature 20.0 °C	0.0000550 ohm-cm @Temperature 68.0 °F	
Thermal Properties	Metric	English	Comments
CTE, linear	10.3 µm/m-°C @Temperature 0.000 - 100 °C	5.72 µin/in-°F @Temperature 32.0 - 212 °F	
	12.2 µm/m-°C @Temperature 0.000 - 650 °C	6.78 µin/in-°F @Temperature 32.0 - 1200 °F	
Specific Heat Capacity	0.460 J/g-°C @Temperature 0.000 - 100 °C	0.110 BTU/lb-°F @Temperature 32.0 - 212 °F	
Thermal Conductivity	25.1 W/m-K @Temperature 27.0 °C	174 BTU-in/hr-ft ² -°F @Temperature 80.6 °F	
Melting Point	1455 - 1510 °C	2651 - 2750 °F	
Solidus	1455 °C	2651 °F	
Liquidus	1510 °C	2750 °F	
Maximum Service Temperature, Air	620 °C	1150 °F	Continuous Service
	735 °C	1360 °F	Intermittent

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AISI Type 420F(Se) Stainless Steel, annealed bar

Component Elements Properties	Metric	English	Comments
Carbon, C	>= 0.15 %	>= 0.15 %	
Chromium, Cr	13 %	13 %	
Iron, Fe	84 %	84 %	
Manganese, Mn	<= 1.0 %	<= 1.0 %	
Molybdenum, Mo	<= 0.60 %	<= 0.60 %	Optional
Phosphorus, P	<= 0.040 %	<= 0.040 %	
Selenium, Se	>= 0.15 %	>= 0.15 %	
Silicon, Si	<= 1.0 %	<= 1.0 %	
Sulfur, S	>= 0.030 %	>= 0.030 %	

References for this datasheet.

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